Software Test Plan (STP)

Smart Complaint Management System (SCMS)

Project: Smart Complaint Management System (SCMS)
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1. Introduction

Purpose: This document defines the test plan for the Smart Complaint Management System (SCMS) Version 2.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing, in order to validate all functional, non-functional, security, and external integration requirements defined in the SRS.

Scope: Testing covers SCMS features including citizen registration and RBAC-based authentication, complaint lodging with multimedia evidence upload, lifecycle tracking and audit trail, admin triage and departmental assignment, resolution verification, citizen feedback/rating, automated notifications, and reporting/analytics. Automated physical maintenance dispatch hardware, internal municipal payroll systems, and off-platform legal workflows are excluded.

References: SCMS Software Requirements Specification (SRS) Version 2.0, WCAG 2.1 Level AA guidelines, OWASP Top 10, TLS 1.3 (RFC 8446).

Definitions: SCMS: Smart Complaint Management System. RBAC: Role-Based Access Control. SLA: Service Level Agreement (maximum stipulated time to acknowledge or resolve a grievance). CRUD: Create, Read, Update, Delete. MIME: Multipurpose Internet Mail Extensions (used for secure attachment validation). JWT: JSON Web Token (for stateless authentication). RTM: Requirements Traceability Matrix.

2. Test Items

- Authentication & RBAC module (Citizen, Department Staff, Administrator)
- Complaint Lodging & Multimedia Evidence Upload module
- Lifecycle Tracking & Audit Trail module
- Admin Triage & Department Assignment module
- Resolution Verification module
- Notifications module (SMS / Email)
- Citizen Feedback & Rating module
- Reporting & Analytics / Export module

3. Features to be Tested

Features mapped to SRS requirement IDs:

- SCMS-F-001: Citizen Registration
- SCMS-F-002: Secure Authentication
- SCMS-F-003: Role-Based Access Control
- SCMS-F-004: Account Lockout Protection
- SCMS-F-005: Lodge Grievance Form
- SCMS-F-006: File Evidence Upload
- SCMS-F-007: Unique Tracking ID Generation
- SCMS-F-008: Payload Field Validation
- SCMS-F-009: Public Tracking Search
- SCMS-F-010: Citizen Dashboard
- SCMS-F-011: Audit Status History
- SCMS-F-012: Citizen Append Notes
- SCMS-F-013: Admin Triage Filtering
- SCMS-F-014: Department Assignment
- SCMS-F-015: Resolution Verification
- SCMS-F-016: Rejection Justification
- SCMS-F-017: Automated Notifications
- SCMS-F-018: Citizen Star Rating
- SCMS-F-019: Analytics Data Export
- SCMS-NF-001: Transaction Response ≤ 2.0s
- SCMS-NF-002: 99.5% Service Uptime
- SCMS-NF-003: 500 Concurrent Session Capacity
- SCMS-NF-004: WCAG 2.1 AA Compliance
- SCMS-NF-005: Daily Backup Retention (<30 min restore)
- SCMS-SR-001: TLS 1.3 Transport Security

- SCMS-SR-002: Argon2id / bcrypt Password Hashing
- SCMS-SR-003: Input Sanitization (SQLi / XSS)
- SCMS-SR-004: 30-Minute Idle Session Timeout
- SCMS-SR-005: Magic-Byte File Whitelisting

4. Features Not to be Tested

- Automated physical on-site maintenance dispatch hardware.
- Internal municipal payroll software integrations.
- Off-platform legal litigation workflows.

5. Test Approach / Strategy

Levels:

- Unit tests (module-level, ≥ 80% coverage; pytest for Django, PHPUnit for Laravel)
- Integration tests (DB state transitions, ACID properties, Celery/Redis async queues, external APIs)
- System / End-to-End tests (Selenium/Cypress automation of the full citizen-to-officer lifecycle: Submitted → Assigned → In Progress → Resolved/Rejected)
- Acceptance tests (UAT against SRS exit criteria)

Types:

- Functional testing (core modules)
- Regression testing (across staging builds each phase)
- Performance testing (JMeter response time and load)
- Security testing (OWASP ZAP, Burp Suite, Qualys SSL Labs)
- Usability / accessibility testing (Lighthouse, screen-reader audit, WCAG 2.1 AA)

Entry Criteria: SRS Version 2.0 reviewed, baseline frozen, and approved; staging environment configured with DB migrations, seed data, and external service mocks; CI/CD build deployed with 100% passing unit tests (≥ 80% code coverage).

Exit Criteria: 100% of High-priority functional requirements (SCMS-F-001 to SCMS-F-017) verified as Passed; 0 open S1 (Critical) or S2 (Major) defects and 0 open High/Critical OWASP vulnerabilities; JMeter validates ≤ 2.0s 90th-percentile response time under 200 users with stable throughput at 500 concurrent threads; complete bidirectional traceability confirmed in the RTM.

5.1 Security Validation

- TLS 1.3 verification across all public and staff endpoints (Qualys SSL Labs Grade 'A')
- Argon2id / bcrypt password hashing checks (cost factor ≥ 12)
- OWASP Top 10 vulnerability scanning (SQLi, XSS) via OWASP ZAP, plus manual Burp Suite penetration runs
- 30-minute idle session timeout enforcement
- Magic-byte file inspection to block disguised executable uploads
- Brute-force lockout defense (5 failed attempts → 30-minute account lockout)

- RBAC endpoint boundary protection (HTTP 403 on cross-role access attempts)

6. Test Environment

Hardware: Cloud/on-prem staging server (4 vCPUs, 8 GB RAM, SSD storage); client testing across standard desktop and mobile devices.

Software: Linux Ubuntu 22.04 LTS; Python 3.11+ (Django 5.x) or PHP 8.2+ (Laravel 11.x); PostgreSQL 15+ / MySQL 8.0 Enterprise.

Tools: Apache JMeter (performance), OWASP ZAP and Burp Suite (security), Qualys SSL Labs (TLS configuration), Selenium/Cypress (E2E automation), pytest/PHPUnit (unit tests), Google Lighthouse (accessibility), a defect-tracking tool (e.g., GitHub Issues or Jira).

Test Data: Seeded citizen/staff/admin accounts, sample complaint records across wards and categories, boundary-size evidence files, mock SMS/email sandboxes (Twilio Mock API, MailHog/SendGrid Sandbox), a local/mock OpenStreetMap Nominatim instance, and a MinIO S3-mock bucket.

Target Clients: Google Chrome 110+, Mozilla Firefox 115+, Apple Safari 16+, Microsoft Edge 110+, and mobile viewports (iOS Safari, Android Chrome).

7. Test Schedule

Milestones are organized around four execution phases; exact calendar dates are to be finalized against the course submission calendar.

- Phase 1 – Smoke & Static Analysis: automated linting, static code analysis, critical path smoke tests (registration, basic login).
- Phase 2 – Functional & Integration Execution: TS-AUTH, TS-LODGE, TS-TRIAGE, and TS-NOTIF suites across staging builds.
- Phase 3 – Security & Non-Functional Validation: TS-SEC scans, JMeter load runs (TC-PERF-01, TC-SCALE-01), mobile accessibility testing.
- Phase 4 – UAT & Regression: full RTM verification and end-to-end verification against acceptance criteria.

8. Test Deliverables

- Test Plan (this document)
- Detailed test cases (29 total — functional, non-functional, and security)
- Test scripts (Selenium/Cypress E2E suites, JMeter load scripts, OWASP ZAP scan configs)
- Test data (seed data, sandbox credentials, boundary-case files)
- Test execution logs
- Defect reports
- Test summary report
- Requirements Traceability Matrix (RTM), cross-referenced with SRS §8

9. Roles and Responsibilities

Role	Name	Responsibility
Team Lead	Atharva Singh	Prepare and coordinate the test plan; liaise with the course coordinator.
QA / Test Lead	Aum Padia	Design and execute test cases, maintain the RTM, log defects.
Backend Developer	Atharva D. Kulkarni	Support defect fixes and triage for API, database, and integration modules.
Frontend Developer	Atharva Gouni	Support defect fixes and triage for citizen and staff-facing UI.
Course Coordinator	Prof. Asha Kumari	Approve test results and sign off on submission readiness.

10. Risks and Mitigation

Risk	Mitigation
Delay in staging build delivery	Request early smoke-test builds from the development sub-team.
Third-party sandbox downtime (Twilio / SendGrid / MinIO / Nominatim)	Fall back to local mocks (MailHog, MinIO, local Nominatim instance).
OWASP scan surfaces critical findings late	Run security scans in Phase 3, not only immediately before submission.
Limited infrastructure for 500-concurrent-session load testing	Use a cloud VM with distributed JMeter load generators.
Team availability constraints near academic deadlines	Build buffer time into Phase 4 (UAT & Regression).

11. Assumptions & Dependencies

- SRS Version 2.0 remains the frozen baseline for the duration of testing.
- A staging environment with DB migrations, seed data, and external sandbox mocks is available before execution begins.
- The CI/CD pipeline delivers builds with ≥ 80% unit test coverage prior to test execution.
- Third-party sandbox services (Twilio Mock, SendGrid Sandbox, MinIO, Nominatim) remain available for integration testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The authentication/database service is unavailable, blocking smoke tests.
- More than 20% of High-priority test cases fail during initial smoke runs.

Resume testing if:

- The defect fix has been deployed to staging.
- The sanity test suite passes 100%.

13. Test Case Management & Traceability

The RTM ensures every SRS requirement maps to at least one test case. All 29 SCMS requirements map 1:1 to a test case, per the Requirements Traceability Matrix (RTM) in SRS §8 — for example, SCMS-F-001 (Citizen Registration) → TC-AUTH-01, SCMS-F-005 (Lodge Grievance Form) → TC-LODGE-01, and SCMS-NF-001 (Response Time) → TC-PERF-01.

13.1 Authentication & Access Control (TS-AUTH)

Test ID	SRS Ref	Description / Preconditions	Test Steps	Expected Result	Priority
TC-AUTH-01	SCMS-F-001	Citizen Registration & Duplicate Handling	1. Submit form with valid details. 2. Submit identical email/phone.	Valid registration persists; duplicate inputs are rejected with inline validation.	High
TC-AUTH-02	SCMS-F-002	Credential Authentication & Sessions	1. Log in with valid credentials. 2. Log in with wrong password.	Valid login returns encrypted session/JWT cookie; invalid login shows generic error.	High
TC-AUTH-03	SCMS-F-003	RBAC Endpoint Boundary Protection	Authenticate as Citizen and attempt to access /admin/triage or /api/v1/admin/*.	Request is denied with HTTP 403 Forbidden.	High
TC-AUTH-04	SCMS-F-004	Brute-Force Lockout Defense	Enter incorrect credentials 5 consecutive times for an account.	6th attempt is blocked, account locks for 30 minutes, and alert email is triggered.	High

13.2 Complaint Lodging & Tracking (TS-LODGE / TS-TRACK)

Test ID	SRS Ref	Description / Preconditions	Test Steps	Expected Result	Priority
TC-LODGE-01	SCMS-F-005	Form Submission & State Initialization	Fill complaint form (category, title, description, ward) and submit.	Record created with status Submitted and valid timestamp.	High
TC-LODGE-02	SCMS-F-006	Evidence File Size & Count Limits	1. Upload 6 files. 2. Upload 1 file > 5 MB. 3. Upload 3 valid files (≤ 5 MB).	Limits enforced: >5 files or >5 MB files are blocked before or at upload.	High
TC-LODGE-03	SCMS-F-007	Immutable Tracking ID Format	Submit complaint and inspect generated tracking ID.	Generates sequential ID matching regex ^SCMS-\d{6}-[A-Za-z0-9]{5}$.	High
TC-LODGE-04	SCMS-F-008	Mandatory Field & Payload Sanitization	Submit empty form and inputs with <script> tags.	Submission halts; UI marks mandatory fields red; XSS strings are sanitized.	Medium
TC-TRACK-01	SCMS-F-009	Unauthenticated Public Status Search	Search existing tracking ID from a logged-out session.	Status, ward, and lodge date display correctly; complainant PII is masked.	High
TC-TRACK-02	SCMS-F-010	Citizen Dashboard Aggregation	Authenticate as Citizen A and navigate to dashboard.	Displays active/resolved tickets belonging strictly to Citizen A.	High
TC-TRACK-03	SCMS-F-011	Audit Trail Logging	Transition complaint through workflow states.	Audit table logs previous_status, new_status, actor_id, and timestamp.	High
TC-TRACK-04	SCMS-F-012	Note Append Permissions by State	1. Append comment while Under Review. 2. Append comment while Resolved.	Comment appends when Under Review; action is rejected/disabled when Resolved.	Medium

13.3 Admin Triage, Resolution & Analytics (TS-TRIAGE / TS-NOTIF / TS-REP)

Test ID	SRS Ref	Description / Preconditions	Test Steps	Expected Result	Priority
TC-ADMIN-01	SCMS-F-013	High-Volume Filtering & Sorting	Apply filters (ward, SLA, department) against database populated with 100k records.	Filtered results render in ≤ 1.0 second.	High
TC-ADMIN-02	SCMS-F-014	Department Assignment Workflow	Assign Submitted ticket to a staff officer with an internal note.	Status moves to Assigned; assignee notification is triggered.	High
TC-ADMIN-03	SCMS-F-015	Resolution Evidence Enforcement	1. Resolve with note < 20 chars, no photo. 2. Resolve with note ≥ 20 chars + photo.	Validation blocks case 1; case 2 resolves and marks status Resolved.	High
TC-ADMIN-04	SCMS-F-016	Rejection Handling & Justification	Admin rejects complaint and inputs mandatory justification code.	Status updates to Rejected; email with justification is dispatched.	High
TC-NOTIF-01	SCMS-F-017	Notification Delivery Latency	Trigger status transition (Submitted → Assigned).	SMS and Email alerts are queued and sent within 30 seconds.	High
TC-NOTIF-02	SCMS-F-018	Citizen Rating & Window Expiry	1. Rate resolved complaint on Day 2. 2. Attempt rating on Day 8.	Day 2 rating records successfully; Day 8 rating link is expired/rejected.	Medium
TC-REP-01	SCMS-F-019	Export Performance & Stream Output	Request CSV/PDF report export for 10,000 complaint records.	Document generates and downloads in ≤ 3.0 seconds without memory errors.	Medium

13.4 Non-Functional & Security Testing (TS-PERF / TS-SEC / TS-UX / TS-OPS)

Test ID	SRS Ref	Category	Test Steps	Expected Result	Priority
TC-PERF-01	SCMS-NF-001	Performance	Execute Apache JMeter load test simulating 200 concurrent users browsing and querying.	90th-percentile response time ≤ 2.0 seconds.	High
TC-SCALE-01	SCMS-NF-003	Scalability	Stress test backend with 500 concurrent authenticated sessions.	0% transaction failure rate; server CPU usage remains below 80%.	Medium
TC-UX-01	SCMS-NF-004	Usability / A11y	Run Google Lighthouse and screen-reader audit across citizen forms on desktop/mobile.	Lighthouse Accessibility score ≥ 90; full keyboard navigation enabled.	Medium
TC-OPS-01	SCMS-NF-002	Availability	Simulate 30-day monitor synthetic ping probes to health check endpoints.	Unscheduled downtime ≤ 3.6 hours per month (99.5% uptime).	High
TC-OPS-02	SCMS-NF-005	Disaster Recovery	Trigger automated daily snapshot restore to an isolated staging database.	Complete restore completes and passes data verification in < 30 minutes.	High
TC-SEC-01	SCMS-SR-001	Transport Security	1. Send HTTP request to port 80. 2. Run Qualys SSL Labs scan against domain.	Port 80 redirects (301) to HTTPS port 443; SSL Labs report yields Grade 'A'.	High

Test ID	SRS Ref	Category	Test Steps	Expected Result	Priority
TC-SEC-02	SCMS-SR-002	Credential Hashing	Query database user table directly for stored password fields.	Credentials are salted Argon2id or bcrypt hashes with cost factor ≥ 12.	High
TC-SEC-03	SCMS-SR-003	Vulnerability Scan	Run automated OWASP ZAP vulnerability suite on all public/staff endpoints.	Zero High or Critical severity SQLi/XSS vulnerabilities identified.	High
TC-SEC-04	SCMS-SR-004	Session Invalidation	Leave authenticated session idle for 31 minutes, then trigger an API action.	Request is rejected with HTTP 401/403 and redirected to login.	Medium
TC-SEC-05	SCMS-SR-005	Magic-Byte Inspection	Rename a .php/.sh script to .jpg or .png and attempt file upload.	Upload rejected with HTTP 415 due to byte magic-number mismatch.	High

13.5 Test Coverage Summary

Requirement Category	SRS Req. IDs	Req. Count	Test Suite(s)
Functional	SCMS-F-001 – F-019	19	TS-AUTH, TS-LODGE, TS-TRACK, TS-TRIAGE, TS-NOTIF, TS-REP
Non-Functional	SCMS-NF-001 – NF-005	5	TS-PERF, TS-SCALE, TS-UX, TS-OPS
Security	SCMS-SR-001 – SR-005	5	TS-SEC
Total	—	29	All requirements traced 1:1 to a test case (see SRS §8 RTM)

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed
- Defect density (defects per module)
- Defect aging (time-to-resolution by severity)
- Requirement coverage (RTM completeness)

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
Team Lead		
QA / Test Lead		
Course Coordinator		